

Fundamentals versus Speculation: What Really Drives Spillovers and Informed trading in Futures and Spot markets of Refined Petroleum products?

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Abstract

In commodities market, the theory of storage suggests that the basis or spread varies with changes in fundamental supply and demand factors. We thus expect these fundamentals, as opposed to speculative trading, to be the primary triggers of volatility in the spot market. Because of structural and functional differences in spot and futures markets, production adjustment cost, sign and sign of the basis, we expect nonlinear price convergence mechanism and time varying volatility. We employ nonlinear and asymmetric error correction model with time-varying volatility to characterize the spot and futures price dynamics of gasoline and heating oil. We show that there is unidirectional and time-varying volatility spillover from futures to spot market since informed trading occurs in the futures markets. We find that error correction process is essentially asymmetric, non-linear, and volatility inducing. The basis account for more than 92% of total volatility suggesting that it is the fundamentals and not speculative trading that drive volatility and spillovers. Moreover, futures prices converge to equilibrium relationship when the basis is positive due to absence of production adjustment costs. We can thus make inference that conventional methodologies that disregard these convolutions in the study of time series dynamics of variables that exhibit mutual stochastic trend may be mis-specified. Our results are robust under different regimes.

Introduction

Since the inception of futures markets and trading in 1860, there has been a strong debate among academicians, practitioners, regulators, policy makers and market players regarding the role of commodity futures on volatility of commodity spot prices. This has sometimes elicited government intervention in futures trading. A classic example is what happened during and after the 2008 financial crisis. In May 2008, the price of crude oil in US spiked to a record \$135 per barrel (a 40 % increase). The Commodity Futures Trading Commission (CFTC) investigated whether the rise in prices was resulting from manipulation, fraud and speculative activities by major players (traders and analysts, large banks, trading houses and oil firms) in the oil industry. Until the 2008 financial meltdown, few people knew that 19 major banks had been secretly granted exemption by CFTC from hedging regulations and could have triggered spike in crude oil prices. Even with these investigations, some analysts such as Phil Flynn of PFGBest countered that the demand and supply fundamentals were the culprits of the astronomical crude oil price rise. The increasing reports of excessive speculation, particularly in the crude oil futures in 2008, increased pressure on CFTC to investigate energy futures contract and mitigate speculative activities. By April 2010, CFTC had initiated necessary checks and balances to curb speculative activities and also proposed to limit the number of futures contracts, especially oil futures, which players in the financial markets can hold at any given time.

As Titman (2010) aptly put it, this debate of the role of futures trading on the volatility of spot prices has refused to peter out. Empirical evidence is also mixed. For example; Kocagil (1997) find no evidence that increased speculative activities in the metals futures markets leads to decline in spot price volatility. Irwin and Yoshimaru (1999) study 36 commodity markets and

find no evidence of any relationship between commodity pool trading and volatility on the underlying spot commodity markets. Sanders, Boris and Manfredo (2004) find that returns and speculative trading in the crude oil market are positively related while Bohl and Patrick (2012) finds that speculation in agriculture and energy commodities does increase volatility in spot prices. In a study of five major agricultural commodities, Chatrath and Song (1999) find a significantly negative relationship between price jumps, number of speculative contracts and the number of speculators. Proponents of commodity futures markets such as Stoll and Whaley (1990), Swartz and Laatsch (1991), Chan (1992), Swartz and Szakmary (1994), Koutmos and Tucker (1996), Kappi (1997), Yang et al (2001), Kavussanos (2008) and Bohl and Patrick (2012) argue that futures trading enhance information flow, increases informativeness and the depth of the market, facilitate hedging and transfer of risk, act as an important vehicle of price discovery, and may actually ease volatility of the spot market. These proponents justify their arguments by providing empirical evidence using the lead-lag relationship between spot and futures returns.

Opponents of commodity futures markets such as Franklin (1988), Cone et al (1988), Reyes (1996), Wee and Poshakwale (2004), Spyrou (2005), Bohl and Patrick (2012) and Sehgal, Rajput and Dua (2012) to mention just a few, argue that futures trading have destabilizing effects on spot market. This is because the higher degree of leverage and lower transaction costs make futures market an attractive platform for uninformed traders to engage in speculative activities and increase volatility of the spot market. Specifically, opponents buttress their arguments by investigating how the level of volatility in the commodity spot markets has intensified after financialization (introduction of commodity futures) of the commodities.

We argue and provide evidence that failure to incorporate the long run equilibrium relationship that exists between futures and spot and the nonlinear and asymmetric nature of

underlying error correction process leads to not only myopic view of the influence of futures trading on spot returns volatility but it also results in incorrect statistical inferences and difficulties in interpreting the results of lead-lag results relationship. While the opponents of futures trading have focused on the volatility of spot market post futures trading inception, they have not adequately incorporated nonlinearities, asymmetries and time-variation in volatility of spot returns after introduction of futures trading. This has failed to sufficiently account for structural shifts in volatility of spot return.

In this paper, we employ a unified econometric modeling that intermarries and resolves the statistical glitches associated with the approaches of both proponents and opponents. Specifically, we jointly study the information and volatility impact of futures markets on spot market and provide a comprehensive insight into the subtleties between futures and spot markets. Our econometric methodology also account for economic regularities that are germane to the debate. In addition to the contribution above, we also address three focal issues regarding futures-spot price dynamic behavior usually blithely ignored by past studies.

First, studies by Carnero, Pena and Ruiz (2004), Ng and Pirrong (1996), Ross (1989), Black (1986) and Kyle (1985) imply that informed trading can be inferred from relative persistence of volatility in spot and futures markets. If futures returns exhibit higher persistence of volatility, this would confirm the theory that futures markets are the principal locus of informed trading and also supports the findings by Llorente, et al (2002) whose research reveal that when investors trade on private information, price changes are likely to continue. Given the existence of positive private information, informed investors will buy and drive up the price. However, when the information is not perfect, there will be only a partial price increase that will

continue into the future. Therefore, returns are positively autocorrelated when investors trade on private information

Second, past studies find that speculative activities are the main catalysts of volatility in futures market and spot markets. For example: Ederington and Lee (2002) show that speculators in the futures market, consisting mostly of floor traders, have higher trading frequency than hedgers, mostly refiners. On average, floor traders' turnover 19% their positions daily compared to an average of 9% daily turnover for refiners. This evidence is consistent with the idea that the actions of speculators trigger volatility in futures market which is then transmitted to spot market. However, the theory of storage fronted by Williams and Wright (1991), Fama and French (1988) and French (1986) surmises that it is the level of inventories that drives the volatility of spot and futures returns. The opening inventory level has direct effects on the basis or spread (futures prices less spot prices). If changes in basis, after adjusting for carrying costs and time to expiration, significantly influence spot and futures returns volatility, we can conclude that that it is the fundamental supply and demand factors that influence the inventory and basis, and not speculative activities, that are the chief determinants of volatility.

Third, Ross (1989) argues that in the absence of arbitrage opportunities, the volatility of stock price is equal to the volume of information flow while the variance of stock return is proportional to variance of information flows. This suggests that if futures' trading generates more information which is fully, accurately and instantaneously incorporated in spot prices, then, spot price volatility would increase after the inception of futures trading. In sum, futures' trading increases the amount of information available for trading which subsequently increases the volatility of the spot market. This view is supported by Ng and Pirrong (1996) and Wee and

Poshakwale (2004) who find that information first arrives in futures markets. Our analysis should thus show volatility spillover from futures to spot markets and not vice versa.

In this study, we design and employ an econometric methodology that not only account for both time-variation in volatility and long-run equilibrium association between spot and futures prices but also enable us to explicitly investigate the three dynamics examined above. This unified approach complements and facilitates understanding of volatility effects and informational role of futures trading.

After adjusting the basis for storage costs and interest rate, we document four important pieces of evidence. First, the basis account for more than 97% (92%) of the ARCH terms causing volatility of heating oil (gasoline). The effect of basis is also asymmetric and nonlinear in first and second moments modeling. Granted the well-documented explicit relationship between basis and demand and supply conditions, then, our evidence is consistent with the theory of storage and the hypothesis that it is the fundamentals that primarily cause volatility in spot returns as opposed to speculative activities. Second, for both heating oil and gasoline commodities, volatility is more persistent in the futures market relative to spot markets. This is consistent with the hypothesis that informed trading takes place in the futures market. Third, the futures shocks included in the second moment (variance) spot equations are economically and statistically significant but spot market shocks included in second moment futures equations are insignificant. This is consistent with the hypothesis that information flows from futures to spot market and not vice versa. This also confirms that the low transaction costs and ease of trading of futures enables faster response to new information in the futures markets before the same information is incorporated in the spot market.

Lastly, we find, at least for heating oil and gasoline futures and spot prices which have a common stochastic trend, that the error correction mechanism that restores the equilibrium relationship between futures and spot prices is nonlinear (captured by the size of the basis) and asymmetric (captured by the size of the basis) since both the size and sign of the basis influence the speed of convergence to equilibrium. Therefore, conclusions made from linear error correction models might yield partial perspective of the aforementioned debate..

2. Data on refined heating oil and gasoline

We base our analysis on daily spot and futures prices of gasoline and heating oil for the period January 1st 2007 to February 18th 2013. Gasoline and heating oil futures contracts are traded on the New York Mercantile Exchange. According to Ederington and Lee (2002), heating oil futures market is the oldest and one of the largest essential energy futures markets. The standard heating oil and gasoline contracts call for delivery of 1000 barrels (42000 gallons) of heating oil or gasoline free on board (F.O.B) ex barge at the seller's refinery facility in New York harbor (cash market). We therefore analyze daily spot prices of the heating oil and gasoline delivery at New York harbor and the associated heating oil and gasoline futures prices from New York Mercantile Exchange. We use the Goldman Sachs commodity indices (GSCI) from Chicago Mercantile Exchange (CME).

Heating oil and gasoline are traded in over-the-counter markets. Both sellers and users of heating and gasoline oil (refiners and consumers) use the centralized futures market to hedge against unfavorable future price movements. Specifically, refiners generally take short positions in the futures market to lock-in the selling prices while the counterparties (heating oil users such

as airline firms) take long positions to lock-in purchase price. This is consistent with hedging theories. However, as Ederington and Lee (2002) points out, the refiners and users also take opposite positions but in general, refiners still end up being net short while users end-up being net long.

Futures contracts have monthly expiration cycles. We thus use the settlement prices of the nearest expiring futures contract. We try to match the relevant spot and futures prices. To this end, the current month spot price is matched with next month futures price since the delivery on a futures contract occurs from the first to seventh business day of the following month. We also account for potential discontinuities that may be caused by switching of contracts. For example: When we switch from April to May contract on 1st May, the futures returns are computed using May 1 and April 30th May delivery futures prices.

In our cointegration analysis, we include dummy variables equal to 1 for contract switch days and zero otherwise. The dummies are insignificant and are suppressed in reporting our results. Consistent with past studies by Nomikos and Andriosopoulos, (2012), Bekiros and Diks (2008), Lanza, Manera and McAleer, (2006) Ewing, Malik and Ozfidan (2002), Copeland (1992) and Baillie and Myers (1991) among others, we use daily returns, $R_{i,t}$, of spot and futures markets

$$R_{i,t} = \ln(P_{i,t} / P_{i,t-1}) \text{ Where } i=\text{spot, futures}$$

3. Model specification and Theoretical Consideration

There are numerous economic considerations that influence the dynamic relationship between spot and futures returns. In this section, we analyze the basis, error correction process and the convenient yield. The following cost of carry equation defines the theoretical relationship between futures and spot prices.

$$F_t = S_t e^{(r_t + w)(T-t)} \quad (1)$$

In equation (1), t is the current period, T is contract expiration date of futures contract, w is a proportional warehousing cost, r_t is the risk free interest rate at time t proxied by 90 day Eurodollar, F_t is the futures price at t and S_t is the spot price at t

Equation (1) suggests that

$$\ln F_t - \ln S_t - r_t(T-t) - w(T-t) = 0 \quad (2)$$

Equation (2) captures the basis or spread (futures prices less spot prices) adjusted for interest and storage. In equilibrium, the basis should be equal to zero. The theory of storage espoused by Working (1949), Tesler (1958), Ng and Pirrong (1996), Zhong, Darrat and Otero (2004) and Hou and Li (2012), among others state that the basis cannot be zero because of convenience yield. According to Ng and Pirrong (1996), Convenience yield is the streams of benefits refiners enjoy when they have inventory at hand to efficiently meet their operational needs. This convenience dictates that refiners are unwilling to sell their inventory in the spot market albeit the spot price might be higher than futures prices (negative basis) or

$$\ln F_t - \ln S_t - r_t(T-t) - w(T-t) < 0$$

According to Fama and French (1988), Tesler (1958) and Working (1948), convenience yield and inventory level have an inverse relationship. This relationship suggests that there is an equilibrium inventory level at which refiners optimize the convenience yield. If inventory is stationary, then convenience yield is also stationary. If we denote the equilibrium inventory and convenience yield as x , we can redefine equation (2) and re-adjust the basis as follows

$$Z_t = \ln F_t - \ln S_t - r_t(T-t) - w(T-t) + x(T-t) \quad (3)$$

If spot and futures prices are influenced by inventory level, then, z_t captures short-run divergences from equilibrium price and inventory levels. Equation (3) thus adjusts the basis, z_t

for interest, storage cost and long run convenience yield. The spread, z_t , temporarily shifts from positive to negative values. When inventory falls, current production is reduced and current consumption increased. Convenience yield will rise. When inventory increases, current production is increased and current consumption reduced. Convenience yield will decline. This inventory, production, consumption and convenience yield adjustment process will result in temporary misalignment between spot and futures prices. Both of these prices thus follow an error correction mechanism (ECM) as they constantly adjust towards long-run equilibrium. We model the ECM as follows:

$$\Delta \ln S_t = \alpha_0 + \sum_{i=1}^n \alpha_i \Delta \ln S_{t-i} + \sum_{i=1}^n \beta_i \Delta \ln F_{t-1} + \sum_{i=1}^n \delta_i \Delta y_{t-i} + \theta z_{t-1} + v_t \quad (4)$$

$$\Delta \ln F_t = \gamma_0 + \sum_{i=1}^n \gamma_i \Delta \ln S_{t-i} + \sum_{i=1}^n \phi_i \Delta \ln F_{t-1} + \sum_{i=1}^n \lambda_i \Delta y_{t-i} + \psi z_{t-1} + e_t \quad (5)$$

In equations (4) and (5), v_t and e_t are random disturbances while $-1 \leq \psi \leq 0$ and $1 \geq \theta \geq 0$. To guarantee the model is stationary, we also assert that $\psi - \theta \leq 1$. $\Delta \ln S_t$ and $\Delta \ln F_t$ is the spot and futures return respectively while $\Delta y_{t-i} = r_t(T-t) - r_{t-i}(T-t+i)$.

The custom in the past studies and literature is to use equation (4) and (5) to assess the dynamic relationship between futures and spot markets. However, the two equations are premised on the assumption that the adjustment process is not only linear but also symmetric. However, Granger and Terasvirta (1993), Enders and Granger (1998) and Kapetanios, Shin and Snell (2006) and Bekiros and Diks (2008) suggest in their notable studies that the ECM might be asymmetric and nonlinear. The speed of adjustment is incalculably influenced by the size of the basis, z_t and the sign of the basis (Whether $z_{t-1} > 0$ or $z_{t-1} < 0$). The size and sign of the basis, among other factors, cause nonlinearities and asymmetries in error correction process of spot and

futures prices of heating and gasoline. We elucidate other causes of nonlinearities and asymmetries as follows:

First, when the basis is marginal, refiners or processors have no incentive to adjust production capacity since it would be unprofitable. There is thus a neutral band of inactivity associated with marginal change in basis as documented by Enders (2010). However, when the basis is large, it becomes profitable to adjust production capacity. This will cause rapid adjustment mechanism in futures and spot prices.

Second, production costs are indivisible and producers will thus attempt as much as possible, to profit from economies of scale. When spot prices are abnormally high, it becomes profitable to actuate production capacity. Conversely, when spot prices are abnormally low, capacity will be idled. These cost indivisibilities become vital triggers of nonlinearities in speed of adjustment mechanism.

Third, when the convenient yield is positively large, producers are expected to rapidly and significantly expand their output to profit from a comparatively high spot price. However, the marginal costs of such adjustment are prohibitively high because of capacity expansion constraints. Capacity or output reduction constraints are minimal when convenience yield is negative or abnormally low. The differential capacity adjustment costs will cause nonlinear speed of adjustment to equilibrium prices. We model these possibilities using equations (6) and (7)

$$\Delta \ln F_t = \alpha_0 + \sum_{i=1}^n \alpha_i \Delta \ln S_{t-i} + \sum_{i=1}^n \beta_i \Delta \ln F_{t-i} + \sum_{i=1}^n \delta_i \Delta y_{t-i} + \theta_1 z_{t-1}^+ + \theta_2 z_{t-1}^- + \theta_3 (z_{t-1}^+)^2 + \theta_4 (z_{t-1}^-)^2 + \eta_t \quad (6)$$

$$\Delta \ln S_t = \gamma_0 + \sum_{i=1}^n \gamma_i \Delta \ln S_{t-i} + \sum_{i=1}^n \phi_i \Delta \ln F_{t-i} + \sum_{i=1}^n \lambda_i \Delta y_{t-i} + \psi_1 z_{t-1}^+ + \psi_2 z_{t-1}^- + \psi_3 (z_{t-1}^+)^2 + \psi_4 (z_{t-1}^-)^2 + \xi_t \quad (7)$$

In equations (6) and (7), $z_{t-1}^+ = z_{t-1}$ if $z_{t-1} > 0$ and $z_{t-1}^- = 0$ if $z_{t-1} \leq 0$, and $z_{t-1}^- = z_{t-1}$ if $z_{t-1} < 0$ and $z_{t-1}^+ = 0$ if $z_{t-1} \geq 0$. The parameters, θ_1 and θ_3 (ψ_1 and ψ_3) quantify the speed of adjustment of the futures (spot) price to positive adjusted spreads. By the same token, θ_2 and θ_4 (ψ_2 and ψ_4) measure the speed of adjustment of the futures (spot) prices to negative adjusted spreads. We include the quadratic terms ($(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$) to capture more rapid convergence when the basis or spreads are large (size and hence nonlinearity) while z_{t-1}^+ and z_{t-1}^- capture the sign of the basis and hence asymmetry. η_t and ξ_t are random error terms. We explain the relationship between speed of convergence and production adjustment costs as follows:

Spot and futures prices will converge at different speeds depending on the sign of the basis (Whether $z_{t-1} > 0$ or $z_{t-1} < 0$). The production adjustment costs associated with the spot prices will mean that futures prices, which have no adjustment costs, are likely to experience higher speed of adjustment when basis is positive, $z_{t-1} > 0$ than when it is negative ($z_{t-1} < 0$). This will show up in our estimate if, $|\theta_1| < |\theta_2|$, $|\theta_3| < |\theta_4|$, $|\psi_1| < |\psi_2|$ and $|\psi_3| < |\psi_4|$. Similarly, if a combination of arbitrage and production causes the futures to converge more quickly whenever $|z_{t-1}|$ is large, then we expect that $|\theta_1| > |\theta_2|$, $|\theta_1| < |\theta_3|$, $|\theta_2| < |\theta_4|$, $|\psi_1| < |\psi_3|$ and $|\psi_2| < |\psi_4|$.

This analysis and interpretation enables us to identify the direction of information flow between futures and spot markets. Informed traders use their information to trade either in the spot or futures markets. If trading occurs in the futures (spot) market, prices will respond to new information and subsequently cause price changes in the spot (futures) market. The market where price response occurs first is the platform of price discovery. The prices of the market in which trading occurs will respond to the information and the price changes will then be transmitted to the next market. The low transaction costs, ease of trading and low start-up costs in futures

markets make the market attractive for price discovery leadership. If β_i and γ_i coefficients in equations (6) and (7) respectively are statistically significant, then we have evidence consistent with information flow. However, information flow may be gradual due to the structure and price setting process of each market.²

4. Informed trading, asymmetric information spillover and time-varying volatility in spot and futures returns

The residuals η_t and ξ_t from equations (6) and (7) respectively should exhibit asymmetric and time-varying volatility for three main reasons: First, Tauchen and Pitts (1983) and Ross (1989) argue that increase in volatility is preceded by increase in rate of flow of information. The modeling by Kyle (1985) also shows trading based on private information should cause persistence in volatility shocks due to gradual incorporation of private information into prices. Cao, Coval and Hirshleifer (2002) also argue that Informed traders in the futures markets may be delaying their trade until price movements validate their private signals. Trading should thus generate delayed internal arrival of further news in the market.

Second, there is no prior evidence that positive and negative basis, caused by high and low inventory levels respectively, have symmetric effects on volatility. We argue that negative basis caused by dearth of supply have more pronounced effects on volatility than copious supply. This will cause asymmetry on the effects of positive and negative basis on volatility hence volatility will vary as the sign of the basis change over time.

Third, Brennan (1991) and Ng and Pirrong (1996) argue that volatility is magnified by short-run inventory supply constraints (which causes negative basis) due to limited adjustment

² We expect spot prices to respond to new information more slowly because of higher transaction costs and the complex nature of the price-setting process. This is unlike the futures contracts that are traded on the over-the-counter markets in a continuous auction.

ability. In the long-run, supply is largely elastic (basis is positive) due to ability to adjust production. In sum, the larger the decrease in basis z_t (and the larger z_t is in absolute values), the higher the volatility since futures and spot prices fluctuate more when supply is fairly inelastic in the short-run. Volatility will vary over time with significant difference between short-run and long-run volatility.

We use a bivariate asymmetric error correction exponential GARCH (BAEC-EGARCH) to capture the three issues

$$\ln h_{f,t} = \omega + \lambda \ln h_{f,t-1} + \mu_1 \left[\frac{|\varepsilon_{f,t-1}|}{\sqrt{h_{f,t-1}}} - E \frac{|\varepsilon_{f,t-1}|}{\sqrt{h_{f,t-1}}} \right] + \mu_2 \frac{\varepsilon_{h,t-1}}{\sqrt{h_{f,t-1}}} + \mu_3 \frac{\xi_{s,t-1}^+}{\sqrt{h_{s,t-1}}} + \mu_4 \frac{\xi_{s,t-1}^-}{\sqrt{h_{s,t-1}}} + \mu_5 (z_{t-1}^+)^2 + \mu_6 (z_{t-1}^-)^2 \quad (8)$$

$$\ln h_{s,t} = \sigma + \kappa \ln h_{s,t-1} + \nu_1 \left[\frac{|\varepsilon_{s,t-1}|}{\sqrt{h_{s,t-1}}} - E \frac{|\varepsilon_{s,t-1}|}{\sqrt{h_{s,t-1}}} \right] + \nu_2 \frac{\varepsilon_{s,t-1}}{\sqrt{h_{s,t-1}}} + \nu_3 \frac{\eta_{f,t-1}^+}{\sqrt{h_{f,t-1}}} + \nu_4 \frac{\eta_{f,t-1}^-}{\sqrt{h_{f,t-1}}} + \nu_5 (z_{t-1}^+)^2 + \nu_6 (z_{t-1}^-)^2 \quad (9)$$

In equations (8) and (9), $h_{f,t}$ and $h_{s,t}$ are the conditional variances of $v_{f,t}$ and $v_{s,t}$ respectively³. The conditioning of variances is based on all information available up to and including $t-1$.

$$\begin{pmatrix} \hat{v}_{f,t} \\ \hat{v}_{s,t} \end{pmatrix} \Omega_{t-1} \sim GED \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} h_{f,t} & \sigma_{s,f,t} \\ \sigma_{s,f,t} & h_{s,t} \end{pmatrix}, \Phi \right),$$

Where $\sigma_{s,f,t} = \rho \sqrt{h_{f,t} h_{s,t}}$ and Ω_{t-1} is the information available as of time $t-1$, and Φ is the inverse of the number of freedom in the GED-distribution. Nelson (1991) argues that GED can

³ To account for potential seasonality of heating oil and gasoline commodities, we detrend returns by regressing the futures and spot returns on time trend and extract the estimated residuals $\hat{v}_{f,t}$ and $\hat{v}_{s,t}$. We use the detrended returns to compute the variances $h_{f,t}$ and $h_{s,t}$

be used in presence of kurtosis and ‘fat-tail’ distribution of security returns. We assume dynamic correlation between spot and futures residuals.⁴

The parameter $\lambda(\kappa)$ is a measure of the persistence of futures (spot) return volatility shocks. The larger λ and κ are, the longer the persistence of volatility shocks. Therefore, λ and κ can be construed as important indicators of informed trading in the futures and spot markets respectively. Carnero, Pena and Ruiz (2004) argue that accurate measurement of persistence of volatility is important because it has implications for the theoretical models used to evaluate the impact of financial shocks and in determination of informed trading.

Trading in both futures and spot markets takes place due to information. Information may flow between the two markets⁵. A shock in spot (futures) market may intensify volatility in the futures (spot) market. Moreover, positive and negative shocks may induce volatility in asymmetric manner. We model these possibilities by including ξ_{t-1}^+ and ξ_{t-1}^- (η_{t-1}^+ and η_{t-1}^-) in the futures (spot) return variance (second moment) equations to capture volatility spillovers between spot and futures markets.

The inclusion of $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ in the second moment equations (8) and (9) have important implications. First, we allow basis to induce volatility in each market in asymmetric manner. Second, preclusion of $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ from second moment equations means that any estimates of spillovers and persistence of volatility are not only biased and incomplete but we would also be drawing incorrect inferences about the dynamics of volatility spillover and the nature of information flow. Third, if we are to somehow confirm the assertion of theory of

⁴ We test if the covariance between futures and spot returns are time-varying by regressing the product of the residuals η_t and ξ_t from equation (6) and (7) on the product's own 10 lags. The joint F-statistic of 10.122 for gasoline and 1.7 for heating oil leads to rejection of the null hypothesis of constant covariance.

⁵ Zhong, Darrat and Otero (2004) find volatility spillover between Mexican equity futures and spot prices.

storage that the prime causes of volatility are fundamental supply and demand forces as opposed to speculative activities, the relative contribution of the ARCH terms in the second moment equations can expose useful information towards that goal.

We estimate equations (6), (7), (8) and (9) jointly to improve the efficiency of parameter estimates. The joint estimation of the parameters account for the form of heteroskedasticity and make our results less conservative in interpretation of nonlinearity and asymmetry of error correction terms and more complete in conclusion regarding direction of information flow. In the OLS estimation of equations (6) and (7), we use 10 lags to purge potential autocorrelation in residuals. In equations (8) and (9), we allow the vector of residuals η_t and ξ_t from equations (6) and (7) to follow a two-dimensional Gaussian Bivariate asymmetric Error Correction EGARCH of order one [BAEC-EGARCH(1,1)] process and bear conditional generalized error distribution (GED) with zero-mean and a bivariate conditional variance-covariance matrix

$$\varepsilon_t = \begin{bmatrix} \eta_t \\ \delta_t \end{bmatrix} \sim GED\left(0, \sum_t \Omega_{t-1}\right)$$

Where $\sum_t = \begin{bmatrix} \sigma_{1,t}^2 & \\ \sigma_{12,t} & \sigma_{1,t}^2 \end{bmatrix}$ and the information set Ω_{t-1} contains past information on lagged basis or error correction term, asymmetric error correction from first-moment equations (6) and (7) and BAEC-EGARCH second-moment equations (8) and (9). As opposed to past studies which use normal distribution, we use GED to account for fat-tails in joint distribution of η_t and ξ_t in estimation of time-varying volatility. The Bivariate EGARCH model also improves parameter estimations. The parameters of our modified bivariate EGARCH model is estimated using numerical maximization and algorithm of Berndt, Hall, Hall, and Hausman (BHHH, 1974). We do not impose any restrictions on the parameter.

From the forgoing, we make a compendium of three testable hypotheses: First, informed traders, using their private signals to trade, causes persistence in volatility in the futures markets. Volatility then spills over to spot market. Second, spot-futures basis or spread depends on the cost of altering production and consumption. The size and sign of the basis will vary with adjustment costs. The variation in sign and size of basis will trigger asymmetry and nonlinearity in the speed of error correction towards convergence of spot and futures prices. Third, the dynamic relationship among basis, elasticity of supply, stock levels and even storage constraints point to potential asymmetric effects of basis on volatility.

5. Empirical Analysis: Preliminary Evidence

We first test whether futures prices, spot prices, yield (90-day Eurodollar interest rate) and the basis exhibit unit root behavior. We use both the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests with a drift. The PP test is used because it is robust to both autocorrelation in residuals and heteroskedasticity⁶. Using 5% significance level and critical value of -2.57, results in table 1 show that we fail to reject the null of unit root for futures and spot prices.

Table (i): Unit root tests and unrestricted cointegration regression

Panel A: Unit root tests

Variable	Heating oil		Gasoline	
	ADF (constant)	PP(constant)	ADF(constant)	PP(constant)
Futures prices	-1.6309	-1.6335	-2.0293	-1.9065
Spot prices	-1.5955	-1.6107	-2.1404	-2.0022
Yield(90-day Eurodollar)	***-9.1034	***-29.5510	***-9.1034	***-29.5510
Basis-one	***-5.7947	***-22.0406	***-6.6853	***-7.5668
Basis-two	***-5.7916	***-22.0170	***-6.6888	***-7.5714

Panel B: unrestricted cointegration analysis

Regressor	Heating oil		Gasoline	
	Coefficient	t-statistic	Coefficient	t-statistic
Ln(Spot prices)	0.9989	3919	0.99966	860.77
yield	4.25E-05	0.17	0.000334	0.304

⁶ Heteroskedasticity is corrected used Newey-West method.

R squared

0.999

0.977

However, we decisively reject the null of unit root for yield, basis-one and basis-two⁷ implying that both are stationary. The evidence of stationarity is even stronger when we use PP test. We also run unrestricted cointegration equations for heating oil and gasoline using $\ln F_t$ as the dependent variable according to equation 2. The $\ln(\text{spot prices})$ coefficient explains almost 100% change in $\ln F_t$ (R squared of 0.999 and 0.977 for heating oil and gasoline respectively) while yield is severely insignificant. Our overall results in this section have important implications for subsequent modeling.

First, since both futures and spot prices have unit root but the basis is stationary, we can safely conclude that futures and spot prices for heating oil and gasoline are cointegrated. This finding is consistent with Granger representation theorem of Engle and Granger (1987) which dictate that we should include error correction term in first moment or mean equations (4) and (5). Second, the fact that yield is stationary while spot and futures prices are non-stationary implies that the three series are not cointegrated. We thus suppress any empirical results associated with the yield in our subsequent analysis.⁸

Having established that cointegration is extant, we investigate preliminary evidence from descriptive statistics, autocorrelations and cross-correlations. Table (ii) shows that futures returns for each of the past 6 years (2007-2012) for both heating and gasoline have leptokurtic (fat or heavy-tail distributions with kurtosis greater than 3). Traders thus expect decline in future

⁷ Basis one is the residuals obtained by regressing detrended $\ln F_t$ on $\ln S_t$ and $r_t(T-t)$ while basis two is residuals obtained by regressing detrended $\ln S_t$ on $\ln F_t$ and $r_t(T-t)$

⁸ The theory of storage suggests that it is imperative to include yield in our analysis to adjust for time value of money in the futures-spot dynamic relationship. However, the inclusion and exclusion of the yield have almost zero effect on the final results. Moreover, our focus on this study is the relationship between futures and spot prices and not the yield.

volatility of these returns. It could also indicate excessive trading or optimism by traders who may be more informed. There is also a higher chance of extreme unfavorable outcomes.

Heating oil (gasoline) spot returns have thin tail or platykurtic distribution in 2010 (2007) with kurtosis less than 3 but fat-tail distributions for all other years. Thin tailed distribution may be indicative of trader pessimism in the market or potential increase in future volatility.

Though the skew of distribution changes from positive to negative, the overall results indicate negative skewness or distribution. Cao, Coval and Hirshleifer (2002) find that informed traders in the futures markets may be delaying their trade until price movements validate their

Table (ii): Yearly descriptive statistics for futures and spot returns

Panel A: Heating oil		Year	2007	2008	2009	2010	2011	2012	2013	All
Futures return	Mean		0.00402	-0.00473	0.00273	0.00163	0.00144	0.00049	0.00135	0.00094
	Median		0.00200	-0.00300	0.00200	0.00000	0.00000	0.00000	0.00250	0.00000
	S. Dev.		0.03422	0.07517	0.04342	0.03523	0.04876	0.03715	0.02505	0.04739
	Skew.		0.03182	0.16684	-0.10145	0.00542	-0.73968	0.29213	-0.26146	-0.11855
	Kurt.		3.22032	3.71984	3.12760	3.02048	6.18478	4.05081	3.08987	5.83293
Spot returns	Mean		0.00403	-0.00509	0.00305	0.00167	0.00143	0.00049	0.00130	0.00093
	Median		0.00100	-0.00150	0.00200	0.00000	0.00000	0.00000	0.00050	0.00000
	S. Dev.		0.03455	0.07469	0.04227	0.03671	0.04719	0.03702	0.02663	0.04707
	Skew.		-0.03915	0.15386	-0.02839	-0.07367	-0.56280	0.29698	0.16536	-0.09455
	Kurt.		3.83054	3.55765	3.40465	2.90024	5.77365	4.54791	2.51887	5.59842
		Obs.	260	262	261	261	260	261	40	1605
Panel B; Gasoline oil			2007	2008	2009	2010	2011	2012	2013	All
Futures return	Mean		0.0034	-0.0056	0.0040	0.0015	0.0009	0.0005	0.0062	0.0009
	Median		0.0040	0.0000	0.0030	0.0000	0.0010	0.0000	0.0080	0.0010
	S. Dev.		0.0450	0.0750	0.0458	0.0355	0.0614	0.0562	0.0341	0.0543
	Skew.		-0.4224	-0.4085	-0.4491	-0.2856	-0.4650	-0.9517	0.0410	-0.6268
	Kurt.		4.0241	3.9936	4.4049	3.2926	5.8165	17.9693	2.4335	8.1321
Spot returns	Mean		0.0032	-0.0061	0.0044	0.0016	0.0009	0.0005	0.0052	0.0009
	Median		0.0040	-0.0010	0.0050	0.0020	0.0000	0.0000	0.0060	0.0020
	S. Dev.		0.0464	0.0849	0.0452	0.0402	0.0570	0.0589	0.0372	0.0570
	Skew.		-0.1454	-0.3092	-0.1355	-0.1778	-0.2773	-0.2763	-0.0940	-0.3928

Kurt.	2.8966	8.3324	3.4561	4.0945	4.9992	7.7515	2.5179	9.6491
Obs.	260	262	261	261	260	261	40	1605

private signals. Trading should thus generate internal arrival of further news in the market. This causes negative return skew following gradual price increases and positive return skew after price declines. It could also cause either disconnect between arrival of new information and large price change or amplified volatility after large changes in security prices. Overall, we provide evidence of non-normal distribution of returns (positive kurtosis and left-tail distribution) and time variation in volatility of heating oil and gasoline returns. The existence of kurtosis may be indicative of potential volatility clustering in the futures and spot returns. Volatility is time-varying since the standard deviation of returns varies each year. For heating oil, futures returns have higher overall volatility than spot returns. This is reversed in the gasoline market where spot returns have higher volatility.

Table (iii): Cross-correlations

Cross-correlation of current futures (spot) returns and lagged spot (futures) returns ΔF_t and ΔS_{t-1} (ΔF_{t-1} and ΔS_t) and cross-correlation of current squared futures (spot) and lagged squared spot (futures) returns ($(\Delta F_t)^2$ and $(\Delta S_{t-1})^2$ [$(\Delta F_{t-1})^2$ and $(\Delta S_t)^2$])

Panel A: Heating oil					Cross autocorrelations					
Lags	1	2	3	4	5	6	7	8	9	10
ΔF_t and ΔS_{t-1}	0.016	-0.018	0.025	0.043	-0.013	-0.016	0.028	0.002	-0.011	-0.004
ΔF_{t-1} and ΔS_t	-0.005	-0.02	0.009	0.052	-0.022	0.001	0.017	0.003	-0.021	-0.009
$(\Delta F_t)^2$ and $(\Delta S_{t-1})^2$	0.156	0.164	0.151	0.145	0.193	0.171	0.052	0.082	0.201	0.173
$(\Delta F_{t-1})^2$ and $(\Delta S_t)^2$	0.125	0.134	0.164	0.123	0.157	0.166	0.071	0.098	0.155	0.165
Panel B: Gasoline oil										
ΔF_t and ΔS_{t-1}	0.001	-0.038	-0.036	0.041	0.032	0.008	-0.01	0.02	0.000	0.043
ΔF_{t-1} and ΔS_t	-0.01	-0.06	-0.017	0.059	-0.008	-0.003	-0.01	0.004	-0.003	0.033
$(\Delta F_t)^2$ and $(\Delta S_{t-1})^2$	0.205	0.174	0.095	0.066	0.069	0.155	0.119	0.026	0.078	0.076
$(\Delta F_{t-1})^2$ and $(\Delta S_t)^2$	0.103	0.099	0.174	0.099	0.051	0.02	0.023	0.094	0.105	0.103

For both heating oil and gasoline, the cross-correlations between ΔF_t and ΔS_{t-1} and $(\Delta F_t)^2$ and $(\Delta S_{t-1})^2$ are generally higher than cross-correlations between ΔF_{t-1} and ΔS_t and between $(\Delta F_{t-1})^2$ and $(\Delta S_t)^2$. For example: For heating oil (gasoline), the cross-correlation between $(\Delta F_t)^2$ and $(\Delta S_{t-1})^2$ is higher than cross-correlation between $(\Delta F_{t-1})^2$ and $(\Delta S_t)^2$ in eight (five) out of ten lags. These preliminary results on cross-correlations support the hypothesis that futures prices lead spot prices in price discovery process

5.1. Empirical evidence from first moment equations

In heating oil market, the futures (spot) mean return equation shows the past eight (three) out of ten lagged spot (futures) returns influence changes in futures (spot) returns. In gasoline market, the futures (spot) regression equations show that the past six (eight) out of ten lagged spot (futures) returns influence changes in futures (spot) returns.

Table (iv): Heating oil Mean equations

LB(4), LB(8) and LB(16) are P-values of Ljung-Box statistics at 4, 8 and 16 lags which test the null of no autocorrelation in residuals. Both LB²(8) (the p-value of Ljung-Box test of squared residuals) and ARCH(8) tests the null of no autoregressive conditional heteroskedasticity for 8 lags.

Regressors	Futures			Spot		
	Coefficient	z-Stat	Prob.	Coefficient	z-Stat	Prob.
Constant	0.004	3.259	0.001	0.004	2.812	0.005
ΔF_{t-1}	-0.339	-6.038	0.000	0.137	2.280	0.023
ΔF_{t-2}	-0.389	-5.960	0.000	-0.181	-3.429	0.001
ΔF_{t-3}	-0.372	-5.424	0.000	-0.251	-3.720	0.000
ΔF_{t-4}	-0.005	-0.077	0.939	-0.021	-0.325	0.745
ΔF_{t-5}	-0.257	-3.161	0.002	-0.111	-1.449	0.147
ΔF_{t-6}	-0.120	-1.547	0.122	-0.105	-1.506	0.132
ΔF_{t-7}	-0.211	-2.785	0.005	-0.076	-0.998	0.318
ΔF_{t-8}	-0.190	-2.503	0.012	0.002	0.019	0.985
ΔF_{t-9}	-0.188	-2.713	0.007	-0.069	-0.964	0.335
ΔF_{t-10}	-0.096	-1.928	0.054	-0.086	-1.476	0.140
ΔS_{t-1}	0.344	6.215	0.000	-0.123	-2.200	0.028
ΔS_{t-2}	0.384	5.969	0.000	0.186	3.601	0.000
ΔS_{t-3}	0.383	5.409	0.000	0.269	4.067	0.000
ΔS_{t-4}	0.058	0.861	0.389	0.041	0.619	0.536
ΔS_{t-5}	0.242	2.912	0.004	0.104	1.360	0.174

ΔS_{t-6}	0.116	1.461	0.144	0.122	1.716	0.086
ΔS_{t-7}	0.230	3.093	0.002	0.105	1.337	0.181
ΔS_{t-8}	0.199	2.478	0.013	-0.009	-0.118	0.906
ΔS_{t-9}	0.201	2.758	0.006	0.057	0.783	0.434
ΔS_{t-10}	0.103	1.987	0.047	0.080	1.297	0.195
Z_{t-1}^+	-0.576	-3.434	0.001	-0.211	-1.267	0.205
Z_{t-1}^-	0.387	2.884	0.004	0.394	2.976	0.003
Z_{t-1}^{+2}	5.049	1.412	0.158	2.666	0.776	0.438
Z_{t-1}^{-2}	5.965	2.597	0.009	5.371	2.333	0.020
$Z_{t-1}^+ = Z_{t-1}^-$ (F-Stat)		14.824	0.000		4.917	0.027
$Z_{t-1}^{+2} = Z_{t-1}^{-2}$ (F-Stat)		0.046	0.830		0.757	0.385
$Z_{t-1}^+ = Z_{t-1}^- = Z_{t-1}^{+2} = Z_{t-1}^{-2}$		10.627	0.000		3.277	0.011
Diagnostics						
LB(4)			0.331			1.000
LB(8)			0.049			1.000
LB(16)			0.093			0.994
LB ² (8)			0.194			0.931
ARCH(8)		1.509	0.149		0.386	0.929

This evidence implies bidirectional return shocks spillover between futures and spot markets. We investigate the influence of basis in explaining mean returns of futures and spot prices. For heating oil, the coefficients of Z_{t-1}^+ , Z_{t-1}^- and Z_{t-1}^{+2} for futures returns and the coefficient of Z_{t-1}^- and Z_{t-1}^{+2} for spot returns are statistically significant. For Gasoline, the coefficients of Z_{t-1}^- , and Z_{t-1}^{+2} are significant for futures returns while coefficient of Z_{t-1}^{+2} is significant for spot returns. Moreover, the hypothesis $Z_{t-1}^{+2} = Z_{t-1}^{-2}$ is rejected albeit marginally, at 10% significance level (p-value=0.0931).

We can thus make two conclusions from this analysis. First, there is bidirectional returns shock spillovers between futures and spot returns. Therefore, there is information flow between the two markets. Second, not only does basis influence the returns of both futures and spot returns but the sign and size of the basis does matter. Moreover, the F-statistics for the hypothesis $Z_{t-1}^+ = Z_{t-1}^-$ is

decisively rejected for both futures and spot returns. With exception of gasoline spot returns in table (v), results in table (iv) and (v) reveal that $|z_{t-1}^+| < |z_{t-1}^-|$ and $|(z_{t-1}^+)^2| < |(z_{t-1}^-)^2|$. This implies that futures prices converge to equilibrium faster when the basis is positive (z_{t-1}^+). This is partly because no production adjustment costs are associated with futures trading. Moreover, futures markets are more liquid, low funding requirements and lower transaction costs.

This evidence supports the role of production costs, size and sign of the basis in the speed of convergence of either the futures or spot prices towards equilibrium relationship. Past studies (See for example, Tse and Xiang 2005 and Azizan et al 2007) that have ignored asymmetries and nonlinearities of the basis in cointegration relationships, at least in heating oil and gasoline prices, may have provided flawed estimates and incorrect inferences.

Table (v): Gasoline mean equation

LB(4), LB(8) and LB(16) are P-values of Ljung-Box statistics at 4, 8 and 16 lags which test the null of no autocorrelation in residuals. Both $LB^2(8)$ (the p-value of Ljung-Box test of squared residuals) and ARCH(8) tests the null of no autoregressive conditional heteroskedasticity for 8 lags.

	Futures			Spot		
	Coefficient	z-Statistic	Prob.	Coefficient	z-Statistic	Prob.
Constant	0.0005	0.2577	0.7967	0.0029	1.3055	0.1917
$\Delta \ln F_{t-1}$	-0.0858	-3.0680	0.0022	-0.0659	-1.9757	0.0482
$\Delta \ln F_{t-2}$	-0.1403	-4.1471	0.0000	-0.1123	-3.2944	0.0010
$\Delta \ln F_{t-3}$	-0.1057	-3.2353	0.0012	-0.0843	-2.8798	0.0040
$\Delta \ln F_{t-4}$	-0.0931	-3.0525	0.0023	-0.0359	-0.9631	0.3355
$\Delta \ln F_{t-5}$	-0.0961	-2.8136	0.0049	-0.1208	-3.4469	0.0006
$\Delta \ln F_{t-6}$	-0.0315	-0.7844	0.4328	-0.0547	-1.5144	0.1299
$\Delta \ln F_{t-7}$	-0.0302	-0.7219	0.4703	-0.0577	-1.8668	0.0619
$\Delta \ln F_{t-8}$	-0.1134	-3.1653	0.0015	-0.1592	-4.4287	0.0000
$\Delta \ln F_{t-9}$	-0.0669	-1.8040	0.0712	-0.0701	-1.8919	0.0585
$\Delta \ln F_{t-10}$	-0.0251	-0.6427	0.5204	-0.0592	-1.9192	0.0550
$\Delta \ln S_{t-1}$	0.0705	2.4945	0.0126	0.0523	1.5151	0.1297
$\Delta \ln S_{t-2}$	0.0881	2.8616	0.0042	0.0733	2.1311	0.0331
$\Delta \ln S_{t-3}$	0.0888	2.7598	0.0058	0.0540	1.8265	0.0678
$\Delta \ln S_{t-4}$	0.1424	4.7930	0.0000	0.0843	2.3420	0.0192
$\Delta \ln S_{t-5}$	0.0921	2.7652	0.0057	0.1191	3.5825	0.0003

$\Delta \ln S_{t-6}$	0.0215	0.5636	0.5730	0.0556	1.5747	0.1153
$\Delta \ln S_{t-7}$	0.0132	0.3254	0.7449	0.0504	1.6781	0.0933
$\Delta \ln S_{t-8}$	0.1315	3.8586	0.0001	0.1417	4.0590	0.0000
$\Delta \ln S_{t-9}$	0.0476	1.3317	0.1829	0.0472	1.3106	0.1900
$\Delta \ln S_{t-10}$	0.0553	1.4901	0.1362	0.0685	2.1848	0.0289
Z_{t-1}^+	0.0814	1.1234	0.2613	-0.0410	-0.6075	0.5435
Z_{t-1}^-	-0.1143	-2.6352	0.0084	0.0576	0.9339	0.3503
$(Z_{t-1}^+)^2$	-0.3214	-0.5571	0.5774	0.7892	1.8129	0.0699
$(Z_{t-1}^-)^2$	-0.5000	-3.0856	0.0020	0.0376	0.1159	0.9078
$Z_{t-1}^+ = Z_{t-1}^-$		3.6477	0.0563		0.7469	0.3876
$(Z_{t-1}^+)^2 = (Z_{t-1}^-)^2$		0.1030	0.7483		2.6016	0.0931
$Z_{t-1}^+ = Z_{t-1}^- = (Z_{t-1}^+)^2 = (Z_{t-1}^-)^2$		2.0892	0.0799		4.2835	0.0019
Diagnostics						
LB(4)			0.6070			0.9090
LB(8)			0.7210			0.9960
LB(12)			0.7940			0.9690
LB ² (8)			0.7910			0.2630
ARCH(8): F-Stat		0.4740	0.8750		1.6360	0.1099

5.2. Empirical evidence from second moment equations

We turn our attention to importance of basis in explaining volatility. In futures markets of heating oil in table v, panel A, the coefficient of $(z_{t-1}^+)^2, \mu_5$ is significant (p-value=0.0051) while for the gasoline, the coefficient of $(z_{t-1}^-)^2, \mu_6$ is significant (p-value=0.0098). In the spot return volatility in table vi panel B, both $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ are significant for heating oil (p-value=0.0091 and 0.0821 respectively) while only coefficient of $(z_{t-1}^-)^2, \nu_6$ for gasoline is significant. We test for the nonlinear effect of basis on the volatility of futures. We are able to identify distinct difference between the coefficients of $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$.

Specifically, we test the hypotheses $\mu_5 = \mu_6$ and $\nu_5 = \nu_6$ (Panel A and B of table vi). In both cases, we strongly reject (p-value=0.0167 and 0.0302 respectively) equality of

$(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ for heating oil but we get exact reverse results for gasoline market. Specifically, we fail to reject the test of nonlinearity or asymmetry of the basis for both futures and spot volatility equations. The p-values for $\mu_5 = \mu_6$ and $\nu_5 = \nu_6$ is 0.783 and 0.6039. These contrasting results in the behavior of the basis or error correction term is partly attributable to the use of deliverable spot prices in heating oil and non-deliverable spot prices in gasoline. Turning to asymmetry of the basis, we find that for both heating oil and gasoline, the coefficients of z_{t-1}^+ , ν_3 are significant (P-values of 0.0753 and 0.0057 for heating oil and gasoline respectively). This evidence is consistent with the hypothesis traders in both markets will increase trading frequency (and hence volatility) when the basis is large when production adjustment benefits exceed the associated costs. This finding reinforce our findings in table (iv) and (v) where futures prices converge to equilibrium when basis is large due to nonattendance of production adjustment costs in futures markets.

Table (vi): Information spillover and informed trading

Panel A						
	Heating oil futures			Gasoline futures		
variable	Coefficient	z-Stat	Prob.	Coefficient	z-Stat	Prob.
ω	-0.46232	-4.04665	0.00010	-0.44065	-3.96611	0.00010
μ_1	0.14821	4.35641	0.00000	0.16206	4.58114	0.00000
μ_2	-0.05551	-2.44837	0.01440	-0.00881	-0.42553	0.67050
λ	0.97652	59.18192	0.00000	0.94852	57.51179	0.00000
μ_3	-0.00680	-0.49075	0.62360	0.00262	0.90571	0.36510
μ_4	0.00153	0.49204	0.62270	-0.00013	-0.04920	0.96080
μ_5	3.56862	2.80213	0.00510	0.76973	0.68061	0.49610
μ_6	0.64272	1.51804	0.12900	1.05742	2.58356	0.00980
$\mu_3 = \mu_4$ (F - Stat)		0.33600	0.56240		0.55400	0.45700
$\mu_5 = \mu_6$ (F - Stat)		5.74214	0.01670		0.07600	0.78300
$\mu_5 = \mu_6 = 0$ (F - Stat)		4.20221	0.0151		3.393	0.0339
Panel B						
	Heating oil spot			Gasoline spot		
variable	coefficient	z-stat	p-value	coefficient	z-stat	p-value
σ	-0.23504	-3.78756	0.00020	-1.14614	-4.49174	0.00000
ν_1	0.09846	3.88522	0.00010	0.19058	4.42335	0.00000

ν_2	-0.02841	-1.62111	0.10500	-0.06101	-1.96645	0.04920
κ	0.94587	120.677	0.00000	0.83672	20.69253	0.00000
ν_3	0.00207	1.77892	0.07530	0.00297	2.76565	0.00570
ν_4	0.00027	0.07681	0.93880	-0.00044	-0.11738	0.90660
ν_5	2.06400	2.60664	0.00910	2.34271	1.15405	0.24850
ν_6	0.42471	1.73842	0.08210	3.34132	3.25989	0.00110
$\nu_3 = \nu_4$		0.24980	0.61730		0.77600	0.37900
$\nu_5 = \nu_6$		4.70460	0.03020		0.26930	0.60390
$\nu_5 = \nu_6 = 0$		3.3096	0.02040		5.31347	0.0050

Overall, the asymmetry and nonlinearity of the basis in the refined petroleum markets, is consistent with the differential costs of constricting and escalating production. It also supports the indivisibility of the production cost. When basis is small, there is a neutral band of inactivity (See Zhong et al 2004 and Enders, 2010) since there is no gain in adjusting production. When the basis (divergence of spot prices from futures prices) is large, oil processors are motivated to adjust production due to higher potential for gain (gain from adjustment exceeds adjustment costs). Large basis are thus corrected faster than small basis. However, we confirm, for both gasoline and heating oil that $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ are jointly not equal to zero (P-value of $\mu_5 = \mu_6 = 0$ and $\nu_5 = \nu_6 = 0$ is 0.0151 and 0.0339 respectively for heating oil and gasoline futures return volatility and 0.0204 and 0.0050 for heating oil and gasoline spot return volatility).

Our findings that the basis significantly influence volatility is consistent with the hypothesis that it is the fundamental forces of demand and supply that cause the volatility and not speculative activities. Moreover, the findings of asymmetry concedes with the theory of storage that state that a negative spread leads to a large convenience yield whenever physical stock of commodity is low relative to expected demand. When stocks are low, supply is inelastic since increase in current consumption cannot be met with future production. The burden of all adjustments to changes in units demanded falls on the spot prices. Other factors constant, this

amplifies spot return volatility when $Z_{t-1} < 0$. Lastly, Brennan (1991) hypothesizes that the positive relationship between $(z_{t-1}^+)^2$ and $h_{s,t}$ is consistent with theory of storage because positive basis is prevalent only when there is a mismatch between level of inventory and storage capacity. Specifically, when inventory level is very high, we have storage constraints and the market have limited flexibility in effecting crucial adjustments to demand and supply shocks. These shocks are instead transmitted to prices which amplify volatility.

We investigate the relative contribution of basis parameters, $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ in explaining volatility in second moment equations (8) and (9). We consider all the ARCH terms in equations (8) and (9) and evaluate the percentage contribution of $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ to total volatility using the following equation:

$$f(z_{t-1}, \eta_{t-1}, \xi_{t-1}) = \frac{g(z_{t-1})}{g(z_{t-1}) + v(\eta_{t-1}, \xi_{t-1})}, \quad \text{Where} \quad g(z_{t-1}) \equiv \hat{\mu}_3(z_{t-1}^+)^2 + \hat{\mu}_4(z_{t-1}^-)^2 \quad \text{and}$$

$v(\eta_{t-1}, \xi_{t-1}) \equiv \hat{\mu}_1 \xi_{t-1}^2 + \hat{\mu}_2 \eta_{t-1}^2$. The function $f(\cdot)$ essentially measures the expected volatility shock resulting from $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ relative to the expected volatility innovation resulting from all ARCH terms. We find that in the heating oil futures (spot) return volatility, both $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ contribute 97.97% (97.17%) of total volatility. This compares to an average of 92.14% (97.73%) contribution of both $(z_{t-1}^+)^2$ and $(z_{t-1}^-)^2$ in futures (spot) volatility in the gasoline market. We can thus conclude that the basis explains almost all the ARCH terms or shocks in spot and futures volatility in both heating oil and gasoline markets.

We then investigate the direction of information flow in the futures and spot markets. In the heating oil (gasoline) futures return volatility, we find that neither the positive nor the negative innovations from spot markets or spillover coefficients, μ_3 and μ_4 , is statistically

significant. However, in the heating oil (gasoline) spot return volatility, we find that positive spillover coefficients, v_3 is statistically significant (P-value=0.0753 and 0.0057 for heating oil and gasoline respectively). These findings are consistent with the hypothesis that information flow from futures to spot market and not vice versa. Moreover, the fact that only positive shocks or news from futures intensify volatility in the spot markets is consistent with the findings of Cao, Coval and Hirshleifer (2002). The authors argue that informed traders in the futures markets may be delaying their trade until price movements validate their private signals. This could cause either disconnect between arrival of new information and large price change or amplify volatility after big changes in asset prices. Large price increases generate positive shocks in the futures markets which are transmitted to spot market to strengthen volatility in the spot return volatility.

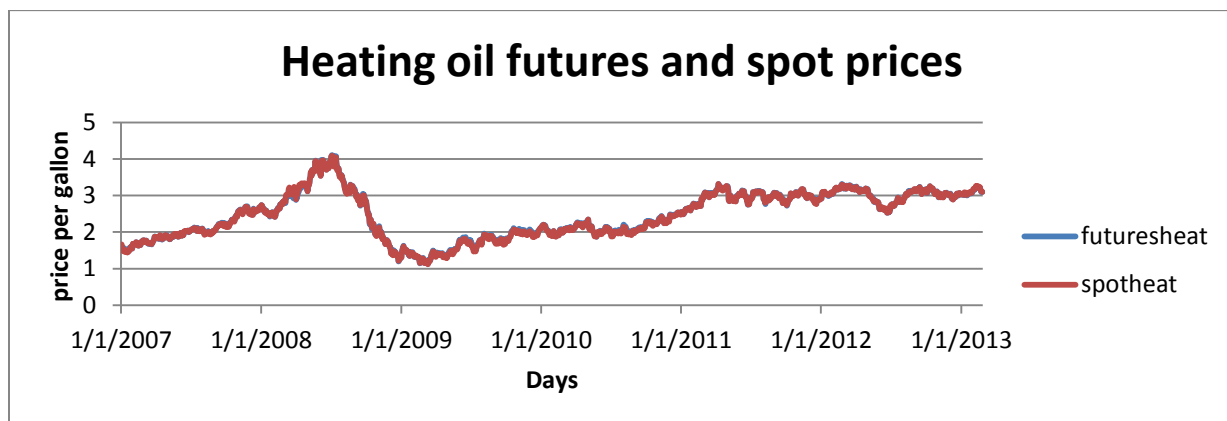
We note that the volatility of spot returns is dependent on past spot volatility $h_{s,t-1}$, and lagged shocks from both spot and futures market. However, the volatility of futures returns does not depend on shocks from spot returns (The parameters μ_3 and μ_4 in table vi panel A are statistically insignificant for both heating oil and gasoline futures). This finding confirms that price discovery, at least in the heating oil and gasoline, mostly takes place in the futures market as information flows from futures market to spot market but not vice versa. Therefore, spot market may have limited role in price discovery process. This finding could be due to higher transaction costs in over-the-counter spot market which makes it difficult and costlier to trade on private information. This contrast to centralized micro-structure of futures market which make dissemination of private information faster. Our finding confirms our preliminary evidence from cross-correlations where correlation between $(\Delta F_t)^2$ and $(\Delta S_{t-1})^2$ is higher than correlation

between $(\Delta F_{t-1})^2$ and $(\Delta S_t)^2$. Our modeling also complements the assertion of Ross (1989) that second moment or volatility equations outperform the first moment equations (returns shocks in VAR analysis) in adequately capturing and incorporating new information in asset prices.

Robustness tests

A visual look (see figure 1) of the futures and spot prices for heating oil reveal potential structural break between in 2008. We use Gregory and Hansen (1996, 1996a, GH hereafter) test for cointegration with regime shifts (structural breaks) using the entire sample period spanning 01/01/2007-02/25/2013. We test for regime shift in the intercept, slope and trend. The GH test is based on the following hypothesis:

Figure 1: Heating oil futures and spot prices



Ho: No cointegration

Ha: Cointegration exists with a single shift (break) at an unknown point in time

For heating oil, structural break occurred on 04/22/2008 while for gasoline, structural break occurred on 10/30/2008. Since GH test can only identify single break, we also test for structural break on the remaining sample after the first break dates. We find no evidence of a structural break.

We jointly re-estimate the mean and conditional variance (equations 6 through 9) under two regimes: Regime 1 spanning 01/01/2007-04/22/2008 (01/01/2007-10/30/2008) for heating oil (gasoline) and regime 2 spanning the period 04/23/2008-02/25/2013 (10/31/2008-02/25/2013) for heating oil (gasoline). To save space, we report estimates of conditional variance equations for both regimes in table (vii).

With exception of regime 1 results for heating oil (panel A), we find regime specific results that are consistent with full sample results. The persistence parameter estimates (λ) are not only significant but estimates for futures are greater than that of spot returns confirming that informed trading takes place in futures markets in different regimes. At least two of the four measures of asymmetry and nonlinearities (μ_3 μ_4 μ_5 and μ_6) are statistically significant. The

Table vii: Robustness tests

Panel A: Heating oil

	Regime 1: 01/01/2007-04/22/2008				Regime 2: 02/23/2008-02-25/2013			
	Futures		Spot		Futures		Spot	
	Coefficient	P-value	Coefficient	P-value	Coefficient	P-value	Coefficient	P-value
ω	-7.5516	0.0000	-8.8953	0.0000	-0.4079	0.0005	-0.5907	0.0005
μ_1	-0.6646	0.0004	0.0349	0.8586	0.1510	0.0001	0.1715	0.0001
μ_2	-0.1462	0.2305	-0.1795	0.1377	-0.0507	0.0436	-0.0520	0.0439
λ	-0.2231	0.2423	-0.2948	0.2283	0.9539	0.0000	0.9300	0.0000
μ_3	-0.2824	0.0309	-0.0034	0.5850	-0.0085	0.5420	0.0082	0.1566
μ_4	0.0202	0.7314	-0.1586	0.0317	0.0014	0.6514	-0.0008	0.8756
μ_5	43.3916	0.0372	73.4604	0.0142	2.5691	0.0538	3.4073	0.0368
μ_6	-63.7136	0.2573	-84.2924	0.2043	0.4884	0.2233	0.8392	0.0723

Panel B: Gasoline

variable	Regime 1: 01/01/2007-10/30/2008				Regime 2: 10/31/2008-02/25/2013			
	Futures		Spot		Futures		Spot	
	Coefficient	P-value	Coefficient	P-value	Coefficient	P-value	Coefficient	P-value
ω	-2.0629	0.0706	-2.6977	0.0332	-0.8826	0.0039	-1.9869	0.0027
μ_1	0.0973	0.3907	0.0918	0.4629	0.2100	0.0001	0.2153	0.0012
μ_2	-0.0512	0.5616	-0.1033	0.2860	-0.0309	0.3243	-0.0823	0.0764
λ	0.6678	0.0004	0.5536	0.0089	0.8830	0.0000	0.7040	0.0000
μ_3	0.0120	0.4747	0.0033	0.0671	-0.0036	0.4987	0.0029	0.0519
μ_4	-0.0056	0.9191	0.0785	0.1122	-0.0011	0.7339	-0.0017	0.7637
μ_5	1.5067	0.1188	2.0149	0.0191	-0.0804	0.9714	-1.8029	0.6562

μ_6	5.6052	0.1931	7.4063	0.0220	1.7278	0.0360	4.7810	0.0139
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size and sign coefficients estimates (μ_1 and μ_2 respectively) of the EGARCH model have the correct sign which indicate that equal magnitude of “bad” and “good” have asymmetric effects on conditional volatility. Therefore, the results presented in table (vi) are largely a persistent feature of the data.

6. Summary and conclusion

This study characterizes the salient features of spot and futures price dynamic behavior of gasoline and heating oil. Using a generalized econometric modeling that attempt to reconcile the case for and against futures trading, we are able to generate numerous important pieces of evidence. First, we are able to demonstrate that information flows or volatility spill over from futures to spot markets. Second, volatility is more persistent in futures market than in spot market which suggests that informed trading occurs in the futures markets. Third, the error correction mechanism is asymmetric and nonlinear whereby the convergence to equilibrium relationship between futures and spot prices is more rapid (sluggish) when absolute basis is large (small) and when the basis is positive (futures prices exceed spot prices). This nonlinear and asymmetric effect of the basis suggest that it is the fundamental forces of demand and supply that precipitate volatility as opposed to speculative activities and noise trading. Fourth and more importantly, the basis or spread account for almost all the innovations or ARCH terms heating oil and gasoline volatility equations.

Our results have important implications for regulators, market participants and researchers. The asymmetric and nonlinear effects of basis on volatility and volatility spillovers suggests that time series analysis of cointegrated variables (short term and long term interest

rates, credit default swaps (CDS) and bond spreads, savings and investments, income and consumption among others) that preclude spreads or basis and its asymmetric and nonlinear effects may yield incorrect estimates and suggest flawed inferences.

As usual, we offer two important caveats in interpreting the results of this study. First, it is possible that our results are driven by excess volatility albeit our evidence is strongly consistent with fundamental-based story of the commodity futures-spot price dynamics. However, nothing in past literature seems to support excess volatility hypothesis. Therefore, we were not able to test this competing hypothesis using our data. Second, our analysis is consistent with Ross (1989) hypothesis that increase in amount of information available to traders in futures market produces information symmetry which subsequently magnifies volatility in spot market. However, Attanasio (1990), Singleton (1987) and Grossman and Stiglitz (1980) offer a competing and contrary hypothesis and argue that less information in the futures market makes the spot prices more volatile due to the information content of spot commodity prices. The authors rationalize that in a market with asymmetric or imperfect information; uninformed traders will intensify their “noise” and prevent prices from fully revealing private signals or information by informed traders. This will magnify volatility in the spot prices. This competing model is neither captured nor tested in this study.

Notwithstanding these caveats, we believe our results have adequately demonstrated and persuasively supported the notion that the basis (which proxy for the fundamentals), explain practically all the ARCH terms or shocks in the heating oil and gasoline volatility equations.

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